

REFLOWECONOMY · REQUIREMENTS

ReflowEconomy requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/refloweconomy		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (15 measurable requirements for the playbook, reference micro-factory and material passport, with status)
0.1	2026-09-24	Amish Chadha	Initial scaffold

ReflowEconomy requirements

These are first-pass requirements for the playbook, its reference micro-factory and the material passport. Targets are proposals for review, not user-validated needs. They will be checked by calculation and source review at TRL 3 and revised after co-design sessions. Status is judged against the concept estimates in RFE-PRC-001; six requirements are **not met** today and four are only partly met.

■ Playbook content

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Say which materials a small operation can recover safely	Feasibility matrix covers at least 7 material groups (plastics, aluminium, steel, glass, lithium cells, e-waste, ore), each with route, feasibility rating, main hazard and export point, and at least one cited source per row	Document review	Partly met: matrix exists with routes and reasons; hazards, export points and sources still to add
R2	Give a process recipe for each locally processed material	One recipe each for PET, HDPE, PP and aluminium: input grade, steps, temperature window, drying, yield, rejects and PPE, each sourced	Document review against published processing data	Not met: no recipes yet
R3	Show whether a micro-factory can pay its way	Economics model per kilogram and per shift with local inputs (feedstock price, wage, energy tariff, product prices) and a break-even output; editable by an operator	Model review with at least one operating recycler	Not met: first-order shift estimate only (RFE-PRC-001), no model
R4	Export only what needs industrial refining	Every export stream names the refining step it needs, the licensed receiver type and the packing rule (for example, cells taped and in a fire-safe container)	Document review	Partly met: principle stated; per-stream rules to write

■ Reference micro-factory

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R5	Fit a small site	Floor area 60 m ² or less (two 40 ft container footprints or a small workshop), with aisles of 1.0 m or more and a separate hot zone	Layout drawing	Met in the massing model (about 60 m ²); aisle widths to check on the drawing
R6	Process a useful amount per shift	100 kg or more of mixed collected input per 8 h shift	Throughput calculation from machine ratings	Met on paper (estimate); shredder and sheet press rates unverified
R7	Keep value local	50 % or more of input mass remanufactured or sold locally as clean feedstock	Mass balance (Figure 2 of RFE-PRC-001), then site data	Met at the limit: about 50 % (estimate)
R8	Limit what goes to disposal	Residue to licensed disposal 20 % or less of input mass; nothing burned in the open	Mass balance, then site data	Met at the limit: about 20 % (estimate); depends on collection quality
R9	Use little energy	1.0 kWh or less per kilogram of output (products plus clean flake)	Energy budget per shift	Met on paper: about 0.9 kWh/kg (estimate)
R10	Stay affordable	Equipment for items 1 to 12 of bom/ bom.csv \$25,000 or less, excluding the building shell	Priced equipment list	Met on paper: about \$22,850 indicative
R11	Protect workers from fumes	Local exhaust on every melt process with a face velocity of 0.5 m/s or more at the hood; no processing of PVC, PS or unknown plastics; workstation air below the local occupational exposure limits	Hood sizing calculation; later air monitoring	Not met: hood not sized; filter type and exposure limits not chosen
R12	Protect workers from machinery, heat, noise and electricity	Guarded shredder with interlocked hopper and emergency stop; insulated hot surfaces; 30 mA RCD on every circuit near water; noise 85 dB(A) or less over 8 h or hearing protection zones	Safety checklist and design review	Not met: shredder noise likely above 85 dB(A), so hearing protection is required; guarding not designed

■ Material passport

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R13	Travel with every lot that leaves	A passport valid against <code>standards/material-passport.schema.json</code> for every product lot, flake bag and export lot	Schema validation of example records	Partly met: schema v0.1 exists; no example records
R14	Trace a lot back to its intake batches	Each passport lists its parent batch IDs and the schema version, so a buyer can follow a product back to its collection points	Schema review	Not met: the schema has no parent batch or version fields (proposed change, awaiting Amish)
R15	Be quick to fill in	A passport completed in 2 min or less per lot on a phone or a shared laptop at the desk	Timed trial with operators	Not verified: needs a form and a trial (TRL 4)

■ Assumptions

- Reference input: source-separated or picker-sorted post-consumer material, of which about 60 % by mass is target plastic (PET, HDPE, PP). Mixed dumpsite material will yield less.
- One 8 h shift a day, 250 shifts a year, four workers.
- PET is sold as clean flake rather than remanufactured in house, because it must be dried thoroughly and degrades easily in small extruders; HDPE and PP become products. This choice is proposed, awaiting Amish.
- Costs are indicative USD for new or locally built equipment and exclude land, building, permits and working capital.